

KITABU QUEST

S6

COMPUTER SCIENCE



Cover scene

students using computers and code safely with a mountain gorilla mascot

● Clear lessons

● Worked examples

● Fun activities

● Real-life problems

GRADE

6

Title Page

KITABU QUEST S6 Computer Science

Rwanda REB-CBC learner book

Mascot: mountain gorilla

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps S6 learners study Computer Science one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Practise safe computing, algorithms, data, systems thinking, and clear step-by-step explanations.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Computer Security
2. Networking Project
3. SQL and Database Project
4. Arrays, functions and procedures
5. Visual Basic project
6. File Management
7. Memory Management
8. Collections in Java
9. Java Enterprise web applications
10. Multimedia
11. File handling in C++

As you finish each topic, return to the success criteria and correct one answer before moving on.

Computer Security: Lesson Opener

Learning focus: Computer Security

Country link: a safe Rwanda observation, model, data table, or investigation for Computer Security.

By the end of this topic, you should be able to:

- enumerate various security threats and ensure security of computers

Inquiry questions:

- How can computer security help you solve a real problem in Computer Science?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Computer Security: Learn

Computer Security is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:

- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:

- Computer Security
- Computer
- Security
- concept
- observation
- evidence
- safety
- conclusion

Computer Security: Worked Example

Worked example for Computer Security: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Computer Security: Vocabulary And Study Skill

Use these words and study moves before you practise Computer Security. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Computer Security
- Computer
- Security
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Computer Security without a clear example, method, or evidence.

Computer Security: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Computer Security.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Computer Security: Practice And Correction

Practice for Computer Security:

Main outcome: enumerate various security threats and ensure security of computers.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Computer Security: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Computer Security.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Computer Security: Outcome Check

Outcome check for Computer Security: Enumerate various security threats and ensure security of computers.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Networking Project: Lesson Opener

Learning focus: Networking Project

Country link: a safe Rwanda observation, model, data table, or investigation for Networking Project.

By the end of this topic, you should be able to:

- build a computer wired and wireless network

Inquiry questions:

- How can networking project help you solve a real problem in Computer Science?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Networking Project: Learn

Networking Project is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:

- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:

- Networking Project
- Networking
- Project
- concept
- observation
- evidence
- safety
- conclusion

Networking Project: Worked Example

Worked example for Networking Project: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Networking Project: Vocabulary And Study Skill

Use these words and study moves before you practise Networking Project. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Networking Project
- Networking
- Project
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Networking Project without a clear example, method, or evidence.

Networking Project: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Networking Project.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Networking Project: Practice And Correction

Practice for Networking Project:

Main outcome: build a computer wired and wireless network.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Networking Project: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Networking Project.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Networking Project: Outcome Check

Outcome check for Networking Project: Build a computer wired and wireless network.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

SQL and Database Project: Lesson Opener

Learning focus: SQL and Database Project

Country link: a safe Rwanda observation, model, data table, or investigation for SQL and Database Project.

By the end of this topic, you should be able to:

- apply Structured Query Language in RDBMS and create a short database project

Inquiry questions:

- How can sql and database project help you solve a real problem in Computer Science?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

SQL and Database Project: Learn

SQL and Database Project is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:

- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:

- SQL and Database Project
- Database
- Project
- concept
- observation
- evidence

SQL and Database Project: Worked Example

Worked example for SQL and Database Project: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

SQL and Database Project: Vocabulary And Study Skill

Use these words and study moves before you practise SQL and Database Project. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- SQL and Database Project
- Database
- Project
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about SQL and Database Project without a clear example, method, or evidence.

SQL and Database Project: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for SQL and Database Project.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

SQL and Database Project: Practice And Correction

Practice for SQL and Database Project:

Main outcome: apply Structured Query Language in RDBMS and create a short database project.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

SQL and Database Project: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from SQL and Database Project.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

SQL and Database Project: Outcome Check

Outcome check for SQL and Database Project: Apply Structured Query Language in RDBMS and create a short database project.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Arrays, functions and procedures: Lesson Opener

Learning focus: Arrays, functions and procedures

Country link: a safe Rwanda observation, model, data table, or investigation for Arrays, functions and procedures.

By the end of this topic, you should be able to:

- use array, functions and procedures in Visual Basic program

Inquiry questions:

- How can arrays, functions and procedures help you solve a real problem in Computer Science?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Arrays, functions and procedures: Learn

Arrays, functions and procedures is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:

- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:

- Arrays, functions and procedures
- Arrays
- functions
- procedures
- concept

Arrays, functions and procedures: Worked Example

Worked example for Arrays, functions and procedures: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Arrays, functions and procedures: Vocabulary And Study Skill

Use these words and study moves before you practise Arrays, functions and procedures. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Arrays, functions and procedures
- Arrays
- functions
- procedures
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Arrays, functions and procedures without a clear example, method, or evidence.

Arrays, functions and procedures: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Arrays, functions and procedures.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Arrays, functions and procedures: Practice And Correction

Practice for Arrays, functions and procedures:

Main outcome: use array, functions and procedures in Visual Basic program.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Arrays, functions and procedures: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Arrays, functions and procedures.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Arrays, functions and procedures: Outcome Check

Outcome check for Arrays, functions and procedures: Use array, functions and procedures in Visual Basic program.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Visual Basic project: Lesson Opener

Learning focus: Visual Basic project

Country link: a safe Rwanda observation, model, data table, or investigation for Visual Basic project.

By the end of this topic, you should be able to:

- connect Visual Basic interface to database and create a simple Visual Basic standard desktop applications for a real life situation

Inquiry questions:

- How can visual basic project help you solve a real problem in Computer Science?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Visual Basic project: Learn

Visual Basic project is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:

- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:

- Visual Basic project
- Visual
- Basic
- project
- concept

Visual Basic project: Worked Example

Worked example for Visual Basic project: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Visual Basic project: Vocabulary And Study Skill

Use these words and study moves before you practise Visual Basic project. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Visual Basic project
- Visual
- Basic
- project
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Visual Basic project without a clear example, method, or evidence.

Visual Basic project: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Visual Basic project.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Visual Basic project: Practice And Correction

Practice for Visual Basic project:

Main outcome: connect Visual Basic interface to database and create a simple Visual Basic standard desktop applications for a real life situation.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Visual Basic project: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Visual Basic project.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Visual Basic project: Outcome Check

Outcome check for Visual Basic project: Connect Visual Basic interface to database and create a simple Visual Basic standard desktop applications for a real life situation.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

File Management: Lesson Opener

Learning focus: File Management

Country link: a safe Rwanda observation, model, data table, or investigation for File Management.

By the end of this topic, you should be able to:

- describe role of operating system in file management and explain file management

Inquiry questions:

- How can file management help you solve a real problem in Computer Science?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

File Management: Learn

File Management is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:

- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:

- File Management
- File
- Management
- concept
- observation
- evidence
- safety
- conclusion

File Management: Worked Example

Worked example for File Management: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

File Management: Vocabulary And Study Skill

Use these words and study moves before you practise File Management. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- File Management
- File
- Management
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about File Management without a clear example, method, or evidence.

File Management: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for File Management.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

File Management: Practice And Correction

Practice for File Management:

Main outcome: describe role of operating system in file management and explain file management.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

File Management: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from File Management.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

File Management: Outcome Check

Outcome check for File Management: Describe role of operating system in file management and explain file management.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Memory Management: Lesson Opener

Learning focus: Memory Management

Country link: a safe Rwanda observation, model, data table, or investigation for Memory Management.

By the end of this topic, you should be able to:

- explain the role of operating system in Memory Management

Inquiry questions:

- How can memory management help you solve a real problem in Computer Science?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Memory Management: Learn

Memory Management is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:

- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:

- Memory Management
- Memory
- Management
- concept
- observation
- evidence
- safety
- conclusion

Memory Management: Worked Example

Worked example for Memory Management: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Memory Management: Vocabulary And Study Skill

Use these words and study moves before you practise Memory Management. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Memory Management
- Memory
- Management
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Memory Management without a clear example, method, or evidence.

Memory Management: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Memory Management.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Memory Management: Practice And Correction

Practice for Memory Management:

Main outcome: explain the role of operating system in Memory Management.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Memory Management: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Memory Management.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Memory Management: Outcome Check

Outcome check for Memory Management: Explain the role of operating system in Memory Management.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Collections in Java: Lesson Opener

Learning focus: Collections in Java

Country link: a safe Rwanda observation, model, data table, or investigation for Collections in Java.

By the end of this topic, you should be able to:

- use collections to store data in Java

Inquiry questions:

- How can collections in java help you solve a real problem in Computer Science?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Collections in Java: Learn

Collections in Java is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:

- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:

- Collections in Java
- Collections
- Java
- concept
- observation
- evidence
- safety
- conclusion

Collections in Java: Worked Example

Worked example for Collections in Java: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Collections in Java: Vocabulary And Study Skill

Use these words and study moves before you practise Collections in Java. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Collections in Java
- Collections
- Java
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Collections in Java without a clear example, method, or evidence.

Collections in Java: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Collections in Java.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Collections in Java: Practice And Correction

Practice for Collections in Java:

Main outcome: use collections to store data in Java.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Collections in Java: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Collections in Java.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Collections in Java: Outcome Check

Outcome check for Collections in Java: Use collections to store data in Java.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Java Enterprise web applications: Lesson Opener

Learning focus: Java Enterprise web applications

Country link: a safe Rwanda observation, model, data table, or investigation for Java Enterprise web applications.

By the end of this topic, you should be able to:

- design and run a Java enterprise web application

Inquiry questions:

- How can java enterprise web applications help you solve a real problem in Computer Science?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Java Enterprise web applications: Learn

Java Enterprise web applications is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:

- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:

- Java Enterprise web applications
- Java
- Enterprise
- applications
- concept
- observation
- evidence
- safety

Java Enterprise web applications: Worked Example

Worked example for Java Enterprise web applications: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Java Enterprise web applications: Vocabulary And Study Skill

Use these words and study moves before you practise Java Enterprise web applications. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Java Enterprise web applications
- Java
- Enterprise
- applications
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Java Enterprise web applications without a clear example, method, or evidence.

Java Enterprise web applications: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Java Enterprise web applications.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Java Enterprise web applications: Practice And Correction

Practice for Java Enterprise web applications:

Main outcome: design and run a Java enterprise web application.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Java Enterprise web applications: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Java Enterprise web applications.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Java Enterprise web applications: Outcome Check

Outcome check for Java Enterprise web applications: Design and run a Java enterprise web application.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Multimedia: Lesson Opener

Learning focus: Multimedia

Country link: a safe Rwanda observation, model, data table, or investigation for Multimedia.

By the end of this topic, you should be able to:

- explain the different use of multimedia and interactive multimedia applications and to use multimedia software to create video

Inquiry questions:

- How can multimedia help you solve a real problem in Computer Science?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Multimedia: Learn

Multimedia is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:

- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:

- Multimedia
- concept
- observation
- evidence
- safety
- conclusion

Multimedia: Worked Example

Worked example for Multimedia: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Multimedia: Vocabulary And Study Skill

Use these words and study moves before you practise Multimedia. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Multimedia
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Multimedia without a clear example, method, or evidence.

Multimedia: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Multimedia.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Multimedia: Practice And Correction

Practice for Multimedia:

Main outcome: explain the different use of multimedia and interactive multimedia applications and to use multimedia software to create video.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Multimedia: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Multimedia.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Multimedia: Outcome Check

Outcome check for Multimedia: Explain the different use of multimedia and interactive multimedia applications and to use multimedia software to create video.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

File handling in C++: Lesson Opener

Learning focus: File handling in C++

Country link: a safe Rwanda observation, model, data table, or investigation for File handling in C++.

By the end of this topic, you should be able to:

- open, close, create a data file in C++ and read, write and append data to Files

Inquiry questions:

- How can file handling in c++ help you solve a real problem in Computer Science?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

File handling in C++: Learn

File handling in C++ is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:

- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:

- File handling in C++
- File
- handling
- concept
- observation
- evidence
- safety
- conclusion

File handling in C++: Worked Example

Worked example for File handling in C++: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

File handling in C++: Vocabulary And Study Skill

Use these words and study moves before you practise File handling in C++. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- File handling in C++
- File
- handling
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about File handling in C++ without a clear example, method, or evidence.

File handling in C++: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for File handling in C++.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

File handling in C++: Practice And Correction

Practice for File handling in C++:

Main outcome: open, close, create a data file in C++ and read, write and append data to Files.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

File handling in C++: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from File handling in C++.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

File handling in C++: Outcome Check

Outcome check for File handling in C++: Open, close, create a data file in C++ and read, write and append data to Files.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Computer Security: Review Clinic 1

Use this review clinic to strengthen Computer Science without copying earlier answers.

1. Choose one idea from Computer Security that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Networking Project: Review Clinic 2

Use this review clinic to strengthen Computer Science without copying earlier answers.

1. Choose one idea from Networking Project that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

SQL and Database Project: Review Clinic 3

Use this review clinic to strengthen Computer Science without copying earlier answers.

1. Choose one idea from SQL and Database Project that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Arrays, functions and procedures: Review Clinic 4

Use this review clinic to strengthen Computer Science without copying earlier answers.

1. Choose one idea from Arrays, functions and procedures that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Glossary

Use this glossary to revise important words in Computer Science.

- Computer Security: Explain this term using an example from the book, your school, or your community.
- Networking Project: Explain this term using an example from the book, your school, or your community.
- SQL and Database Project: Explain this term using an example from the book, your school, or your community.
- Arrays, functions and procedures: Explain this term using an example from the book, your school, or your community.
- Visual Basic project: Explain this term using an example from the book, your school, or your community.
- File Management: Explain this term using an example from the book, your school, or your community.
- Memory Management: Explain this term using an example from the book, your school, or your community.
- Collections in Java: Explain this term using an example from the book, your school, or your community.
- Java Enterprise web applications: Explain this term using an example from the book, your school, or your community.
- Multimedia: Explain this term using an example from the book, your school, or your community.
- File handling in C++: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Computer Science answer states the idea, uses correct scientific vocabulary, records observations or data, includes units or labels where needed, and writes a conclusion that follows from evidence. Safety rules always come first. If evidence is missing, repeat the observation safely or explain what evidence would be needed.

Final Project

Create a final project for Computer Science that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.